Motivation: To navigate the complexities of optimization, we must first establish a clear roadmap of the theoretical and algorithmic milestones we will cover.

Primal-Dual Bounds and Branch-and-Bound Algorithm

Context

Academic Presentation / Third Assignment Solutions

Focus Areas

Evaluating primal and dual bounds, LP relaxations, duality theorems, and exact algorithmic resolution via Branch-and-Bound.

Motivation: Before attempting to solve complex combinatorial problems, we must define the mathematical limits of our optimal objective values.

Motivation: Having defined the bounds, we must establish how to measure the distance between them to mathematically guarantee an optimal solution.

Current Section:

Primal Bound

A valid upper limit (in minimization) on the optimal objective value. Obtained by evaluating the objective function at any feasible solution. The true minimum is <= this bound.

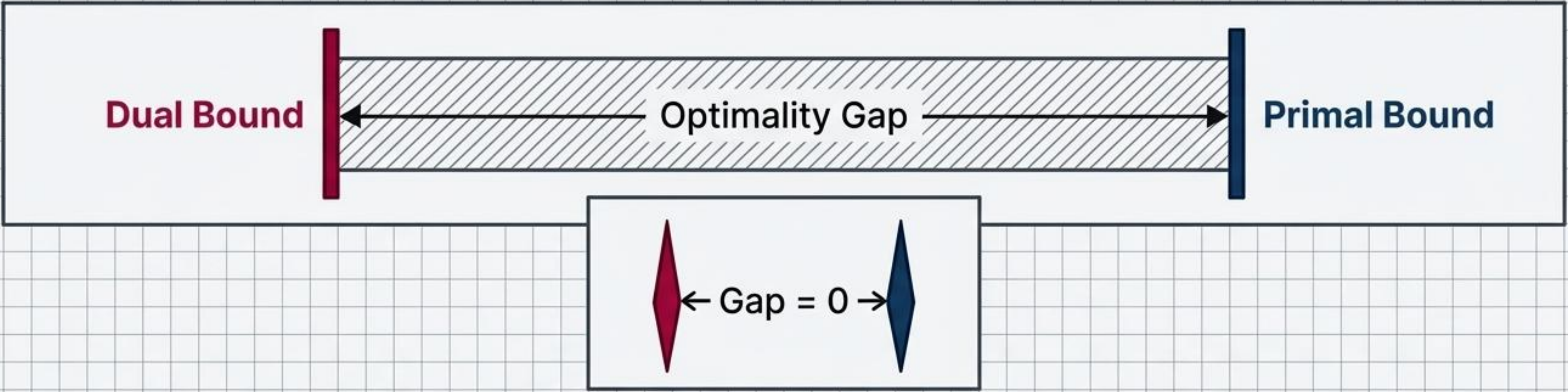
$$(\min z_{opt}) \leq (z_{primal})$$

Dual Bound

A valid lower limit (in minimization) on the optimal objective value. Found via problem relaxation (e.g., ignoring integrality constraints). Always optimistic because relaxations expand the feasible region or remove penalties.

$$(\max z_{dual}) \leq (\min z_{opt})$$

Motivation: To compute these theoretical dual bounds in practice, we frequently employ continuous relaxations of integer constraints.



The Optimality Gap

The absolute or relative difference between the best known primal bound (best feasible solution found) and the best known dual bound (theoretical limit).

Certifying Optimality

Optimality is strictly certified when the Optimality Gap equals zero (Primal Bound = Dual Bound).

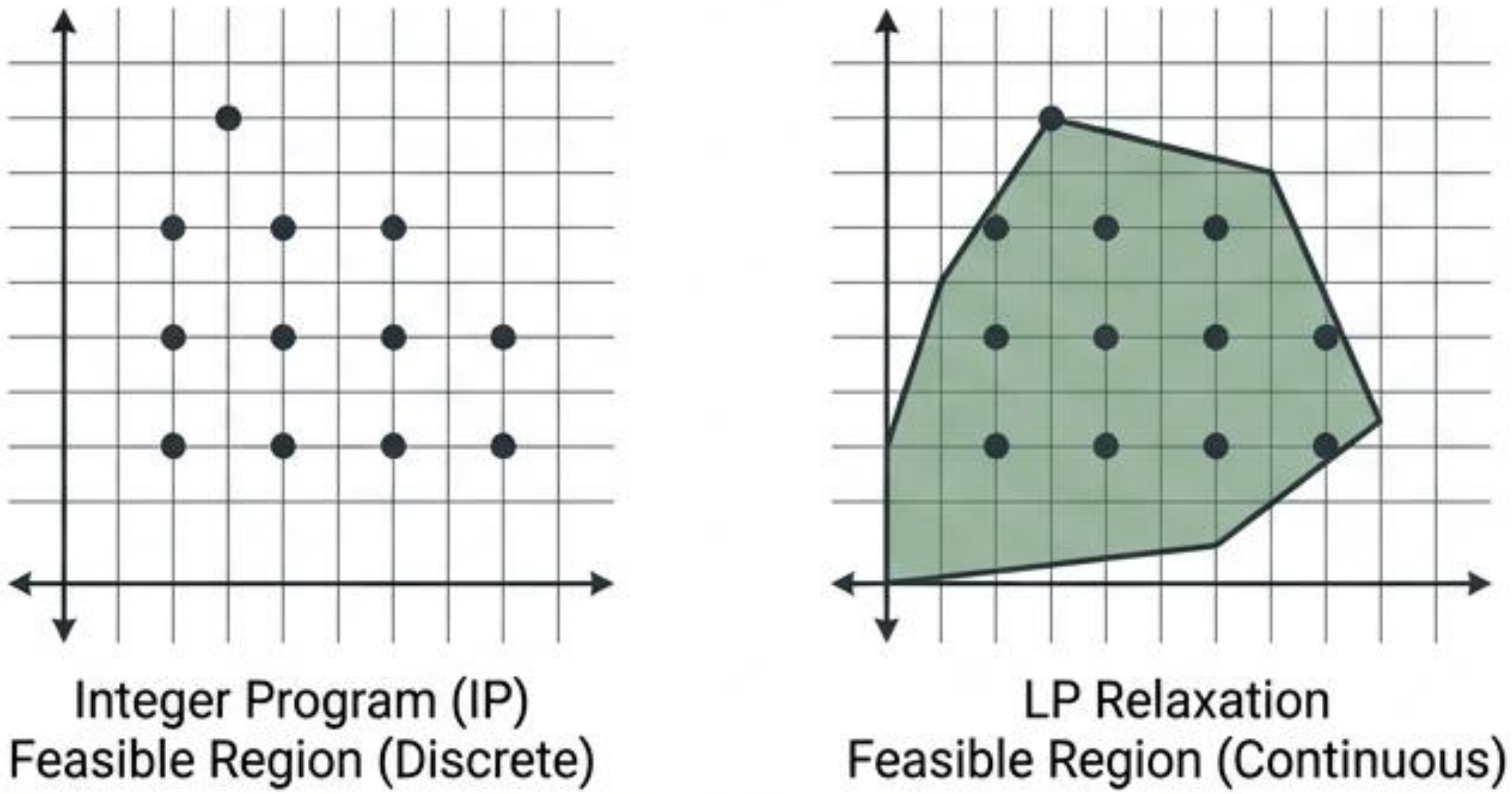
Implication

Proves no better solution can possibly exist within the feasible space.

Motivation: Let us apply this theoretical relaxation to a concrete mathematical model to extract numerical bounds.

Current Section:

LP Relaxation Definition: The linear programming problem obtained by dropping integrality constraints from an Integer Program (IP) or Mixed-Integer Linear Program (MILP).



Mechanism: Allows variables to take continuous fractional values instead of strict integers.

Purpose: Enlarges the feasible region to quickly compute a valid dual bound.

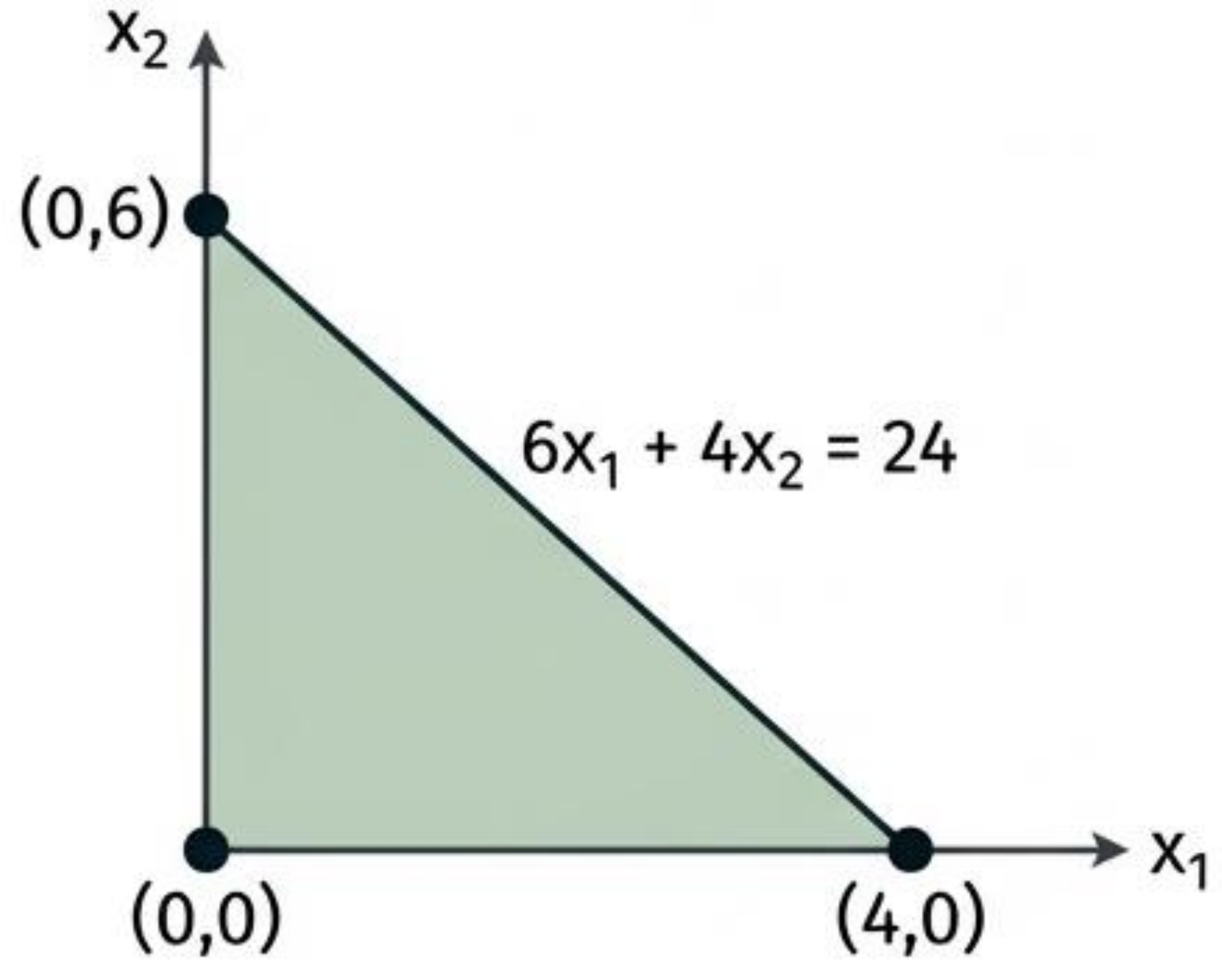
Motivation: The mechanism that guarantees the bounds extracted from relaxations are valid lies in the foundational theories of linear programming duality.

Problem

$$\begin{aligned} \max \quad & 5x_1 + 4x_2 \\ \text{subject to} \quad & 6x_1 + 4x_2 \leq 24 \\ & x_1, x_2 \in \mathbb{Z}_{\geq 0} \end{aligned}$$

Objective

Solve the LP relaxation geometrically to find primal/dual bounds and certify the optimality gap.



Components & Step-by-Step Resolution

1. **Relaxation**: Drop $Z_{\geq 0}$, assume continuous bounds $x_1, x_2 \geq 0$.
2. **Vertices Evaluation**: (0,0) => Z=0 ; (4,0) => Z=20 ; (0,6) => Z=24.
3. **Dual Bound**: Continuous maximum Z_{LP} = 24. (Valid upper bound for maximization).
4. **Primal Bound**: The continuous solution (0,6) consists entirely of integers, making it strictly feasible for the original IP. Primal bound = 24.
5. **Optimality Gap**: $|24 - 24| = 0$. Optimality is mathematically certified.

Motivation: To fully grasp how variables and constraints interact at the point of optimality, we must examine **complementary slackness**.

Current Section:

1. Primal and Dual Bounds
2. LP Relaxation
3. Duality in Linear Programming
4. Bounding Techniques
5. Branch and Bound: An Exact Algorithm

The Dual Problem

Every linear program (primal) has an associated dual. Primal variables become dual constraints; primal constraints become dual variables.

Weak Duality Theorem

The primal objective is bounded by the dual objective (e.g., $c^T x \leq b^T y$ for primal maximization). Any feasible dual solution provides a valid bound.

Strong Duality Theorem

If either problem has a finite optimal solution, both do, and their optimal objective values are strictly equal ($Z^* = W^*$).

Motivation: We will now geometrically solve both the primal and the newly formulated dual to empirically verify the Strong Duality Theorem.

Complementary Slackness

At optimality, strictly satisfied constraints (non-zero slack) imply **paired variables are zero.** Strictly positive variables imply paired constraints are binding (zero slack).

Primal Problem Formulation:

```
max Z = 3x_1 + 2x_2
subject to x_1 + x_2 ≤ 4,
           2x_1 + x_2 ≤ 5,
           x_1, x_2 ≥ 0
```

Objective: Write the dual problem.

Dual Components Formulated:

```
min W = 4y_1 + 5y_2
subject to y_1 + 2y_2 ≥ 3,
           y_1 + y_2 ≥ 2,
           y_1, y_2 ≥ 0
```

Motivation: While LP relaxation is foundational, modern combinatorial optimization requires a broader taxonomy of techniques to generate strong mathematical bounds.

Primal Solve

Intersection of constraints at (1, 3).

Vertices evaluated:

`Z(0,4) = 8, Z(1,3) = 9, Z(2.5,0) = 7.5.`

Maximum Z* = 9 at (x_1=1, x_2=3).

Dual Solve

Intersection of constraints at (1, 1).

Vertices evaluated:

`W(0,2) = 10, W(1,1) = 9, W(3,0) = 12.`

Minimum W* = 9 at (y_1=1, y_2=1).

Strong Duality Verification

The optimal primal objective value (Z* = 9) exactly equals the optimal dual objective value (W* = 9).

Motivation: To observe these bounding techniques computationally, we examine a fractional greedy relaxation of the classic Knapsack Problem.

Current Section:

Strong Bound Definition: A bound very close to the true integer optimal value, creating a tight gap to prune search trees faster.

Approach	Use Case	Bound Quality	Properties
LP Relaxation	IP, MILP	Variable	Fast to compute, drops integrality constraints.
Lagrangian Relaxation	Constrained IP	Strong	Hard constraints moved to objective function and penalized by multipliers.
Surrogate Relaxation	IP	Moderate to Strong	Aggregates multiple constraints into a single weighted row.

Motivation: Bounds and relaxations provide mathematical limits, but to guarantee an exact integer solution, we must systematically divide and conquer the feasible space.

Problem Definition

Problem: $\max \sum_{i=1}^{n} p_i x_i$ subject to $\sum_{i=1}^{n} w_i x_i \leq W, x_i \in \{0, 1\}$.

Components: Profits $p = (10, 8, 6)$, Weights $w = (5, 4, 3)$, Capacity $W = 7$.

Objective: Compute dual bound via fractional knapsack (greedy algorithm) and define integrality gap.

Step-by-step Resolution:

1. Compute Ratios: `10/5 = 2, 8/4 = 2, 6/3 = 2.`
2. Take Item 1 entirely: $w = 5$, $p = 10$. Remaining $W = 2$.
3. Take fraction of Item 2: `2/4 = 0.5` fraction. Profit = `0.5 × 8 = 4.`
4. Total Dual Bound = `10 + 4 = 14.`

Integrality Gap Concept

The absolute difference or ratio between the true optimal objective value of the Integer Program and the optimal value of its continuous LP relaxation.

Motivation: To strictly define this logic, we unify the mathematical formulation, algorithmic sequence, and programmatic instructions into a single framework.

Current Section:

1. Primal and Dual Bounds (Dimmed)

2. LP Relaxation

3. Duality in Linear Programming

4. Bounding Techniques

5. Branch and Bound: An Exact Algorithm

Exact Algorithm Framework: Guarantees finding the absolute mathematical optimum and proving its optimality in finite time. Utilized for strict safety or financial constraints.

Main Idea: 'Divide and Conquer' - Partitioning the feasible region.

Pillar 1: Branching

Systematically partitioning the feasible region into smaller subproblems.

Pillar 2: Bounding

Utilizing relaxations to compute theoretical limits for each subproblem.

Pillar 3: Fathoming (Pruning)

Permanently closing a node if its LP is infeasible, its bound is worse than the incumbent, or its LP solution is already strictly integer.

Motivation: With the exact algorithm fully defined, we apply it to a final mathematical program to observe the bounding and fathoming mechanics in real time.

Mathematical Logic

Formulation: $\max Z \text{ s.t. } x \in S$

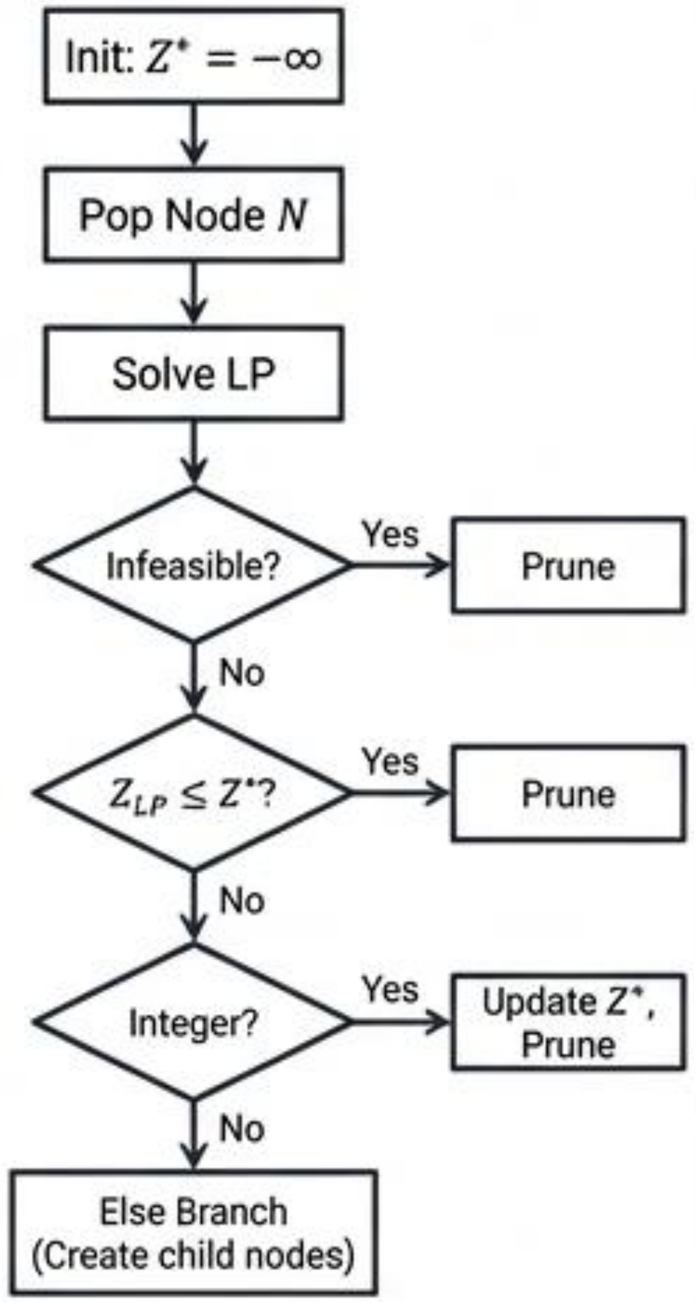
Branching Constraints:

$$x_i \leq \lfloor v \rfloor \text{ and } x_i \geq \lceil v \rceil$$

Optimality Fathoming:

$$Z_{LP} \leq Z^*$$

Flowchart



Structured Pseudocode

```
While Active_Nodes not empty:
    Current = Select(Active_Nodes)
    LP_Sol, LP_Z = Solve_LP(Current)
    If INFEASIBLE or LP_Z ≤ Best_Z: Continue
    If Is_Integer(LP_Sol):
        Best_Z = LP_Z; Continue
    Branch(Floor/Ceil); Add children;
```

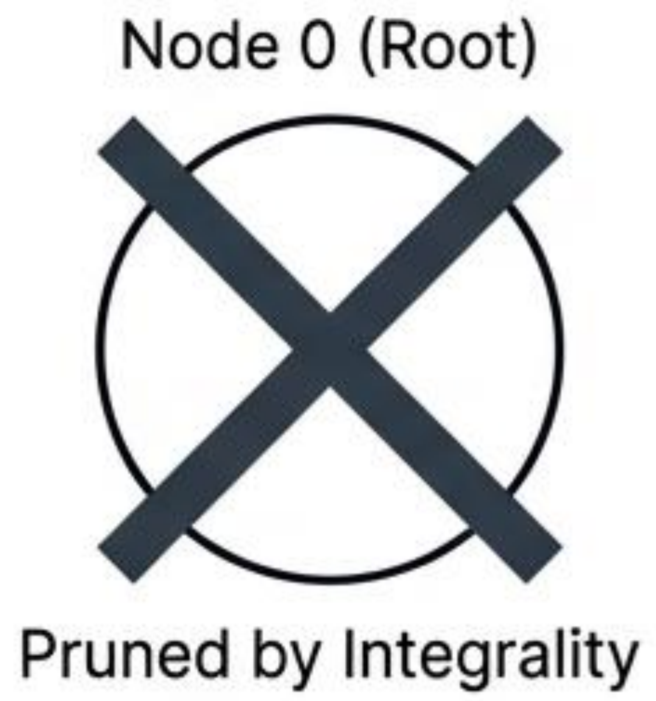
Motivation: Let us review the fundamental theoretical principles established in this presentation.

Problem & Objective

Problem: max 3x_1 + 2x_2 subject to 2x_1 + x_2 ≤ 4, x_1, x_2 ∈ $\mathbb{Z}_{\geq 0}$.

Objective: Execute B&B tree, determine bounds, pruning logic, and optimal solution.

Search Tree Diagram



Components & Step-by-Step Resolution

Node 0 (Root): Solve continuous LP relaxation constraint 2x_1 + x_2 ≤ 4.

Vertices Evaluated: (0,0) ⇒ Z=0; (2,0) ⇒ Z=6; (0,4) ⇒ Z=8.
LP Optimum: Z_{LP} = 8 at (x_1=0, x_2=4).

Fathoming Test: The solution (0,4) is strictly integer. It is directly optimal for the IP.

Pruning Logic: Node 0 is permanently pruned by integrality.

Conclusion: Search tree consists solely of Node 0.
Optimal Solution: x_1=0, x_2=4, Z*=8.

Conclusion: The Mechanics of Exact Optimization

Bounds Dictate Limits

Primal bounds define feasible reality, while dual bounds map theoretical limits; optimality is strictly certified when their gap is zero.

Relaxation Enables Computation

LP, Lagrangian, and Surrogate relaxations temporarily sacrifice integer accuracy to rapidly compute valid dual bounds.

Duality is Symmetric

Strong Duality and Complementary Slackness theorems mathematically guarantee that primal and dual problem spaces mirror one and dual problem spaces mirror one another at optimality.

Branch-and-Bound is Definitive

By unifying branching (division), bounding (relaxation), and fathoming (pruning), B&B ensures an exact integer optimum in finite time.